

Keva Planks

WHAT'S IN THE KIT

- 500 Keva planks per kit
- Brain Builder challenge cards



QUICK-START GUIDE

1. Dump, sort, build — planks stack without connectors.
2. Start with the Brain Builder cards for guided structures.
3. Move to open challenges: tallest tower, longest bridge span.
4. Cleanup tip: stack planks in flat rafts of 10 for fast counting.

FULL LESSONS IN THIS GUIDE

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PRO TIP: 500 planks per kit is enough for 6-8 simultaneous small-group builds.

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LESSON · GRADES K-2

Three Little Pigs Engineering

LEARNING OUTCOMES

- Students will build a stable structure that meets a story requirement
- Students will test a structure against a force and observe results
- Students will describe why structures fall or survive

MATERIALS

- Keva planks (a generous pile per group)
- The Three Little Pigs (any version)
- A 'wolf': small fan or standardized push
- Building mats to define plots

INSTRUCTIONS

1. Read the story; frame the challenge: build a house the wolf can't blow down.
2. Groups build houses for the pigs on their mats.
3. Wolf visits every house — same fan distance/time for fairness.
4. Survivors and fallers examined together: what did survivors share?
5. Rebuild round: everyone applies one 'survivor idea' to their house.
6. Wolf returns. Celebrate improvements, not just survivals.

ACTIVITY

Three Little Pigs Engineering — build, test against the wolf-fan, study the survivors, rebuild. The rebuild round is where actual engineering thinking shows up.

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LESSON · GRADES 3-5

Bridge Span Challenge

LEARNING OUTCOMES

- Students will design a bridge that spans a gap and carries load
- Students will record span and load data across designs
- Students will explain trade-offs between reach and strength

MATERIALS

- Keva planks (equal counts per team)
- Desks or tables to bridge between
- Standard weights (identical books or bags of counters)
- Span/load data table

INSTRUCTIONS

1. Set desks a hand-width apart; every team bridges the gap and holds one weight.
2. Widen the gap in rounds; teams re-engineer each time.
3. Record each round: span, planks used, weight held before collapse.
4. Introduce discoveries as they appear: overlapping, corbeling, triangulation.
5. Final round: teams choose their own max span to attempt — risk vs. glory.
6. Graph class data: span vs. strength. Find the trade-off line.

ACTIVITY

Bridge Span Challenge — an ever-widening gap forces re-engineering every round, and the class data graph makes the span-versus-strength trade-off visible.

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LESSON · GRADES 6-8

Earthquake-Proof Towers

LEARNING OUTCOMES

- Students will engineer a tower to survive lateral shaking
- Students will apply cross-bracing and base design principles
- Students will iterate based on observed failure modes

MATERIALS

- Keva planks
- Shake table (tray on marbles or a wobble board)
- Height requirement marker
- Failure-mode observation sheet

INSTRUCTIONS

1. Set the constraint: minimum height, must survive 10 seconds of standardized shaking.
2. Teams build v1 towers; shake test all of them.
3. Autopsy: where did each tower fail first — base, middle, joints?
4. Teach bracing and base-widening only after failures create the need.
5. Rebuild applying one structural principle; retest.
6. Optional finale: keep shaking until only one tower stands.

ACTIVITY

Earthquake-Proof Towers — shake, fail, diagnose, rebuild. Teaching bracing after the first collapse, not before, is what makes the principle stick.